

Chapter 3: Spaces, Alignment & Pyramids

SLIDE 1 — WHY DO SPACES MATTER?

What Are Spaces in Patterns?

A space is also a character that can be printed as part of a pattern.

Spaces help us control **where a symbol appears** on the screen.

Simple Idea

Spaces → Position → Shape

Left-Aligned Pattern

In a left-aligned pattern, stars begin from the same position.

```
*  
**  
***  
****  
*****
```

Right-Aligned Pattern

In a right-aligned pattern, spaces are printed before the stars.

```
  *  
  **  
 ***  
**  
*****
```

Important Idea

The stars may increase in the same way, but **spaces change their position**.

Remember

Spaces are not empty. They are part of the pattern.

SLIDE 2 — RIGHT-ALIGNED TRIANGLE

What Is a Right-Aligned Triangle?

A right-aligned triangle is a pattern where the stars move toward the right side of the output.

```
*
**
***
**
*****
```

Row	Spaces	Stars
1	4	1
2	3	2
3	2	3
4	1	4
5	0	5

C Example

```
#include <stdio.h>
```

```
int main() {

    int n = 5;

    for (int i = 1; i <= n; i++) {

        for (int j = 1; j <= n - i; j++) {
            printf(" ");
        }

        for (int j = 1; j <= i; j++) {
            printf("*");
        }

        printf("\n");
    }

    return 0;
```

```
}
```

Output

```
*  
**  
***  
**  
*****
```

Important

There are **two inner loops**:

First loop → Prints spaces

Second loop → Prints stars

Remember

Right alignment = Spaces first + Stars after.

SLIDE 3 — SPACE + STAR LOGIC

What Happens in Each Row?

In a right-aligned triangle:

- Spaces **decrease**
- Stars **increase**

This creates the right-aligned shape.

Example

For $n = 5$:

Row 1 → 4 spaces + 1 star
Row 2 → 3 spaces + 2 stars
Row 3 → 2 spaces + 3 stars
Row 4 → 1 space + 4 stars
Row 5 → 0 spaces + 5 stars

The Relationship

For every next row:

Spaces → Decrease by 1

Stars → Increase by 1

Formula

For row i :

Spaces = $n - i$

Stars = i

Example

If:

$n = 5$

$i = 3$

Then:

Spaces = $5 - 3 = 2$

Stars = 3

So row 3 becomes:

Simple Thinking

Row Number



Calculate Spaces



Print Spaces



Calculate Stars



Print Stars



New Line

Remember

Spaces control position. Stars control size.

SLIDE 4 — FULL PYRAMID

What Is a Full Pyramid?

A full pyramid has stars centered around the middle of the output.

```
*
***
*****
*****
*****
*****
```

Row	Spaces	Stars
1	4	1
2	3	3
3	2	5
4	1	7
5	0	9

What Do We Notice?

Spaces decrease:

$4 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0$

Stars increase by **2**:

$1 \rightarrow 3 \rightarrow 5 \rightarrow 7 \rightarrow 9$

C Example

```
#include <stdio.h>
```

```
int main() {
```

```
    int n = 5;
```

```

for (int i = 1; i <= n; i++) {

    for (int j = 1; j <= n - i; j++) {
        printf(" ");
    }

    for (int j = 1; j <= 2 * i - 1; j++) {
        printf("*");
    }

    printf("\n");
}

return 0;
}

```

Output

```

*
***
*****
*****
*****

```

Important

A pyramid needs:

Spaces + Stars

The number of stars is not simply equal to the row number.

Remember

Pyramid = Decreasing spaces + Increasing odd number of stars.

SLIDE 5 — PYRAMID FORMULAS

How Do We Find the Number of Spaces?

For every row, the number of spaces decreases by one.

Row 1 → $n - 1$ spaces

Row 2 $\rightarrow$ $n - 2$ spaces

Row 3 $\rightarrow$ $n - 3$ spaces

So:

Spaces = $n - i$

How Do We Find the Number of Stars?

The stars follow odd numbers:

$1 \rightarrow 3 \rightarrow 5 \rightarrow 7 \rightarrow 9$

The formula for row i is:

Stars = $2 * i - 1$

Example

Suppose:

$n = 5$

$i = 3$

Then:

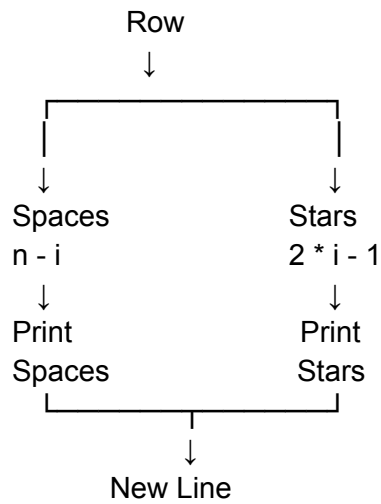
Spaces = $5 - 3$
= 2

And:

Stars = $2 * 3 - 1$
= 5

So row 3 is:

Complete Logic



Final Pattern

```
*
***
*****
*****
*****
```

Remember

Right Triangle:

Spaces = $n - i$

Stars = i

Full Pyramid:

Spaces = $n - i$

Stars = $2 * i - 1$

One-Line Recap

Spaces decide where the pattern starts, while **stars** decide how wide the pattern becomes.

Question

. What is the main purpose of spaces in a pattern?

- A) To increase the number of stars
- B) To control where symbols appear on the screen
- C) To stop the loop
- D) To print numbers

 **Hint:** Think about how spaces affect the position of stars.

 **Glossary:** Position → The place where something appears.

 **Answer:** B) To control where symbols appear on the screen

2. What will be the output of the following C program?

```
#include <stdio.h>
```

```
int main() {  
  
    int n = 4;  
  
    for (int i = 1; i <= n; i++) {  
  
        for (int j = 1; j <= n - i; j++) {  
            printf(" ");  
        }  
  
        for (int j = 1; j <= i; j++) {  
            printf("*");  
        }  
  
        printf("\n");  
    }  
  
    return 0;  
}
```

A)

```
*  
**  
***  
****
```

B)

```
*
**
***
****
```

C)

```
****
***
**
*
```

D)

```
*
***
*****
*****
```



Hint: The first inner loop prints $n - i$ spaces before the stars.



Glossary: Right-aligned → A pattern where spaces are printed before the symbols so they appear toward the right.



Answer: B)

Output:

```
*
**
***
****
```

3. Complete the code to create a right-aligned triangle.

```
int n = 5;
```

```
for (int i = 1; i <= n; i++) {
```

```
    for (int j = 1; j <= _____; j++) {
        printf(" ");
    }
```

```
    for (int j = 1; j <= i; j++) {
        printf("*");
```

```
}  
  
    printf("\n");  
}
```

 **Hint:** For a right-aligned triangle, spaces decrease as the row number increases. The formula is $n - i$.

 **Glossary:** $n \rightarrow$ Total number of rows.

$i \rightarrow$ Current row number.

 **Answer:** $n - i$

4. Complete the code to print the correct number of stars in a full pyramid.

```
int n = 5;
```

```
for (int i = 1; i <= n; i++) {  
  
    for (int j = 1; j <= n - i; j++) {  
        printf(" ");  
    }  
  
    for (int j = 1; j <= ____; j++) {  
        printf("*");  
    }  
  
    printf("\n");  
}
```

 **Hint:** The number of stars in each row follows the sequence 1, 3, 5, 7, 9.

 **Glossary:** Odd numbers $\rightarrow$ Numbers such as 1, 3, 5, 7, 9 that increase by 2.

 **Answer:** $2 * i - 1$

5. Complete the code to print the spaces for a full pyramid.

```
int n = 5;
```

```
for (int i = 1; i <= n; i++) {
```

```
for (int j = 1; j <= _____; j++) {  
    printf(" ");  
}  
  
for (int j = 1; j <= 2 * i - 1; j++) {  
    printf("*");  
}  
  
printf("\n");  
}
```

💡 **Hint:** The number of spaces decreases by 1 for every new row.

📖 **Glossary:** Decrease → To become smaller by a given amount.

✅ **Answer:** $n - i$